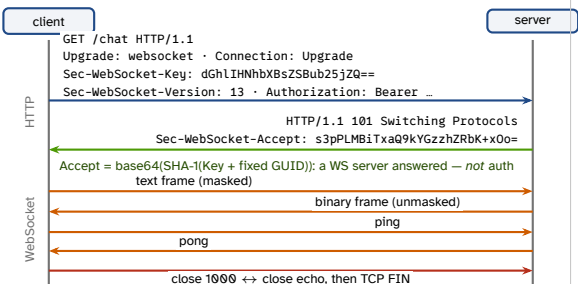


In one line: A **WebSocket** (RFC 6455) is an HTTP/1.1 request that **upgrades** (101) into a full-duplex stream of small **frames** on the same TCP connection. It gives foreground realtime; it does *not* give delivery across reconnects, ordering across connections, or life in the background — you build **heartbeats**, **resubscribe**, **gap fill** and **dedup**, and use **APNs** when the app is not running.

Opening handshake, then frames

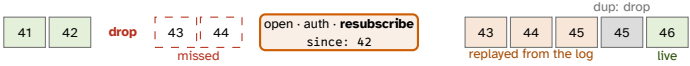


Frame header (2-14 bytes)

F	R	S	V	opcode	MASK	payload len	ext. len	masking key	payload ...
1	1	1	1	4 bit	1	7 bit	0 / 16 / 64	0 / 32 bit	

opcode: 0x1 text (UTF-8) · 0x2 binary · 0x0 continuation · 0x8 close · 0x9 ping · 0xA pong. Length 0-125 inline · 126 → 16-bit · 127 → 64-bit. Client→server frames **must** be masked (against proxy-cache poisoning); control frames ≤ 125 B, never fragmented. Close codes: 1000 normal · 1001 going away · 1002 abnormal (local only, never on the wire).

Reconnect → resubscribe → gap fill → dedup



The server assigns a **seq per channel**; the client persists lastSeq. seq = last + 1 → apply · seq ≤ last → **drop** (duplicate) · seq > last + 1 → **gap**: fetch the range over REST, or re-snapshot if the log no longer has it. SSE has it built in: id: lines + the Last-Event-ID header on reconnect.

Choosing the transport

	direction	latency	battery	infra / catch
polling	pull	= interval	worst: radio per poll	plain HTTP, cacheable
long poll	pull, held	near real-time	a request per batch	held requests, timeouts
SSE	down only	real-time	one stream	auto-resume; proxies buffer
WebSocket	duplex	lowest	foreground only	LB: Upgrade + long idle
APNs	down	seconds, best-effort	best (system link)	throttled; 4 KB payload

Pick the simplest that meets latency: a feed → SSE or polling; chat, collaboration, games → WebSocket; “tell me while I’m not running” → APNs.

How it works

- Auth** happens at the **handshake**: URLSessionWebSocketTask takes a URLRequest, so set Authorization (browsers can’t — they use a query token or a first message). After 101 there are no headers: an expired token needs an in-band re-auth or a reconnect.
- Heartbeats**: NATs silently drop idle mappings; a half-open socket looks *open* until a write fails, minutes later. sendPing every ~20-30 s; no pong in time → dead, reconnect. App-level heartbeats also prove the server *app* lives.
- Reconnect**: exponential backoff + **full jitter**, reset after a stable period; also on NWPathMonitor “satisfied” and on foreground. On open: auth → **resubscribe** (the new server knows nothing) → gap fill → flush outbox.
- Ordering**: TCP orders one connection only; across reconnects, servers and backplanes you need the server seq. Delivery is at-least-once → idempotent handlers, dedupe by message id.
- Background**: iOS suspends the app seconds after it leaves; the socket dies with it. Close cleanly inside beginBackgroundTask; deliver via **APNs**; reconnect + gap fill on foreground. (PushKit VoIP is for calls only.)

Example — URLSessionWebSocketTask (iOS 13)

```
var req = URLRequest(url: URL(string: "wss://api.example.com/ws")!)
req.setValue("Bearer \({token})", forHTTPHeaderField: "Authorization")
let ws = session.webSocketTask(with: req)
ws.resume()
try await ws.send(.string("#{"op":"sub","ch":"room.7","since":42}"))
while true {
    // receive() = ONE message
    // throws when it closes
    switch try await ws.receive() {
    case .string(let s): handle(s)
    case .data(let d): handle(d)
    @unknown default: break
    }
}
// heartbeat timer: ws.sendPing { error in /* nil = pong */ }
// leaving: ws.cancel(with: .goingAway, reason: nil)
```

Server side — fan-out

Sockets are **stateful**: each gateway holds its connections in memory. A message published anywhere must reach sockets on **every gateway** → a **pub/sub backplane** (Redis pub/sub, NATS, Kafka, an MQTT broker): gateways subscribe to the channels their clients want and push down the sockets. The WebSocket itself pins to one gateway, so stickiness only matters for multi-request fallbacks (long polling). Budget per-connection memory; bound each client’s send buffer — a slow consumer gets dropped, not buffered forever. A deploy closes thousands of sockets at once: send close 1001, drain, and rely on client jitter to avoid a reconnect storm.

MQTT over WebSocket

MQTT packets carried in binary frames (Sec-WebSocket-Protocol: mqtt): browsers can’t open raw TCP, and some networks allow only 443. You get topics + wildcards, QoS 0/1/2 (QoS 1 = duplicates → idempotent), retained last value, Last Will — the reconnect/state problems above, already solved by the broker. A native iOS client can use MQTT over TLS directly.

Traps

- ! Not re-arming receive() (completion API): one message, then silence.
- ! “The socket is open, so the peer is alive” — half-open TCP. Heartbeat with a timeout.
- ! Reconnect without **resubscribe** + since: connected, and silently missing messages.
- ! Fixed-delay reconnect: every client returns in the same second after a server blip.
- ! Keeping a socket in the background “with a background mode”: not allowed for this — push.
- ! Sec-WebSocket-Accept is not security; auth the handshake, use wss: // (TLS).
- ! Default maximumMessageSize is 1MB (*unverified*) — a bigger message fails the receive.

Remember

“Upgrade, heartbeat, backoff with jitter, resubscribe, fill the gap, drop the dup — and push when you’re asleep.”